



March 16, 2018

StemSys
Clay Fette
CEO
10871 NW 52nd St, Ste 4
Sunrise, Florida 33351

Re: K172593
Trade/Device Name: XCelliStem Wound Powder
Regulatory Class: Unclassified
Product Code: KGN
Dated: February 13, 2018
Received: February 14, 2018

Dear Clay Fette:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you; however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part 801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820); and if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "*Misbranding by reference to premarket notification*" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <http://www.fda.gov/MedicalDevices/Safety/ReportaProblem/default.htm> for the CDRH's Office of Surveillance and Biometrics/Division of Postmarket Surveillance.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/>) and CDRH Learn (<http://www.fda.gov/Training/CDRHLearn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<http://www.fda.gov/DICE>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

David Krause -S

for Binita S. Ashar, M.D., M.B.A., F.A.C.S.
Director
Division of Surgical Devices
Office of Device Evaluation
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K172593

Device Name

XCELLISTEM WOUND POWDER

Indications for Use (Describe)

The XCelliStem Wound Powder is intended for the management of wounds including: partial and full-thickness wounds, pressure ulcers, diabetic ulcers, venous ulcers, chronic vascular ulcers, tunneled/undermined wounds, surgical wounds (donor sites/grafts, post-Moh's surgery, post-laser surgery, podiatric, wound dehiscence), trauma wounds (abrasions, lacerations, second-degree burns, and skin tears), and draining wounds. The device is intended for one-time use.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

This section applies only to requirements of the Paperwork Reduction Act of 1995.

DO NOT SEND YOUR COMPLETED FORM TO THE PRA STAFF EMAIL ADDRESS BELOW.

The burden time for this collection of information is estimated to average 79 hours per response, including the time to review instructions, search existing data sources, gather and maintain the data needed and complete and review the collection of information. Send comments regarding this burden estimate or any other aspect of this information collection, including suggestions for reducing this burden, to:

Department of Health and Human Services
Food and Drug Administration
Office of Chief Information Officer
Paperwork Reduction Act (PRA) Staff
PRAStaff@fda.hhs.gov

"An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB number."

K172593Applicant:
StemSys**XCelliStem Wound Powder**
Traditional 510(k) Premarket Notification**Section 5**
510(k) Summary**510(k) SUMMARY - XCelliStem Wound Powder**

Company Name: StemSys
 Company Address: 10871 NW 52nd St, Ste 4
 Sunrise, FL 33351

Contact Person: Clay Fette
 Chief Executive Officer

Phone Number: 561-324-9507
 Date Prepared: August 17, 2016

Device Trade Name: XCelliStem Wound Powder
 Device Common Name: Wound Dressing
 Classification Number: Unclassified
 Classification Name: Dressing, Wound, Collagen
 Product Code: KGN

Predicate Devices:

Company	Device	510(k) Number
Cook Biotech Inc.	Cook ECM Powder	K152033
ACell, Inc.	ACell Powder Wound Dressing	K060888

Device Description: The XCelliStem Wound Powder is an extracellular matrix composed of porcine collagen for the management of wounds. It is white to off-white particulate that is to be used as a topical application. The product is provided sterile, for single use.

Intended Use: The XCelliStem Wound Powder is intended for the management of wounds including: partial and full-thickness wounds, pressure ulcers, diabetic ulcers, venous ulcers, chronic vascular ulcers, tunneled/undermined wounds, surgical wounds (donor sites/grafts, post-Moh's surgery, post-laser surgery, podiatric, wound dehiscence), trauma wounds (abrasions, lacerations, second-degree burns, and skin tears), and draining wounds. The device is intended for one-time use.

Substantial Equivalence: XCelliStem Wound Powder has the same intended use, principles of operation, and similar technological characteristics as both predicate devices. While the subject device differs from the predicate devices in the source tissue (porcine spleen and lung vs. porcine small intestine or porcine urinary bladder), all three devices share the same mode of action in regards to covering the wound and providing a

K172593Applicant:
StemSys**XCelliStem Wound Powder**
Traditional 510(k) Premarket Notification

moist wound environment to maintain an appropriate environment for wound management. XCelliStem Wound Powder is as safe and effective and performs as well as the legally marketed predicate devices based on an evaluation of biocompatibility, bench, non-clinical, and clinical performance; any differences in technological characteristics do not raise new questions about the safety and efficacy of XCelliStem Wound Powder.

Summary of Biocompatibility, Non-clinical testing, and Clinical testing

The following testing was performed to demonstrate substantial equivalence:

Biocompatibility testing:

- Cytotoxicity
- Sensitization
- Intracutaneous Reactivity
- Acute Systemic Toxicity
- Pyrogenicity
- Toxicology Risk Assessment
- Implantation
- Genotoxicity

Clinical testing:

- Repeat Insult Patch Test
- Skin Prick Test

Other testing:

- Sterilization Validation
- Viral Inactivation
- Device Characterization: Collagen type composition, Sulfated glycosaminoglycan composition, Hyaluronic acid composition, Lipid composition, Elastin composition, Fibronectin composition, and Laminin composition
- Extractables & Leachables
- Bacterial Endotoxin

K172593

Applicant:
StemSysXCelliStem Wound Powder
Traditional 510(k) Premarket Notification

Comparison Table for Substantial Equivalence

Product Name	Proposed XCelliStem Wound Powder	Cook ECM Powder	ACell Powder Wound Dressing
Product Code, Regulation #, Name	KGN Unclassified Dressing, Wound, Collagen	KGN Unclassified Dressing, Wound, Collagen	KGN Unclassified Dressing, Wound, Collagen
Manufacturer	StemSys	Cook Biotech Inc.	ACell, Inc.
Intended Use	The XCelliStem™ Wound Powder is intended for the management of wounds that include: • Partial and full thickness wounds • Pressure ulcers, venous ulcers, diabetic ulcers, chronic vascular ulcers • Second degree burns • Surgical wounds – donor sites/grafts, post-Moh's surgery, post-laser surgery, podiatric, wound dehiscence • Trauma wounds – abrasions, lacerations, and skin tears • Tunneled/undermined wounds • Draining wounds	Cook ECM Powder is intended for the management of wounds that include: • Partial and full thickness wounds • Pressure ulcers, venous ulcers, diabetic ulcers, chronic vascular ulcers • Second degree burns • Surgical wounds – donor sites/grafts, post-Moh's surgery, post-laser surgery, podiatric, wound dehiscence • Trauma wounds – abrasions, lacerations, and skin tears • Tunneled/undermined wounds • Draining wounds	The ACell™ Powder Wound Dressing is intended for the management of wounds including: • Partial and full thickness wounds • Pressure ulcers, venous ulcers, diabetic ulcers, chronic vascular ulcers • Second degree burns • Surgical wounds – donor sites/grafts, post-Moh's surgery, post-laser surgery, podiatric, wound dehiscence • Trauma wounds – abrasions, lacerations, and skin tears • Tunneled/undermined wounds • Draining wounds
Composition	Collagen type I, III, IV	Collagen type I, III, IV, VI	Collagen types I, III, IV
Collagen Source	Porcine spleen and lung	Porcine small intestine submucosa	Porcine urinary bladder
Sterile	Yes, E Beam	Yes, Ethylene Oxide	Yes, E Beam
How supplied	particulate; for single use	particulate; for single use	particulate; for single use
How applied	Topical use	Topical use	Topical use
Biocompatible	Yes	Yes	Yes
510(k) Number	K172593	K152033	K060888